

A top-down view of a light teal desk. In the top left is a silver laptop with a black keyboard. To its right is a white mouse and a cup of dark coffee. In the bottom left is a brown leather notebook with a white pen and a pair of tortoiseshell glasses.

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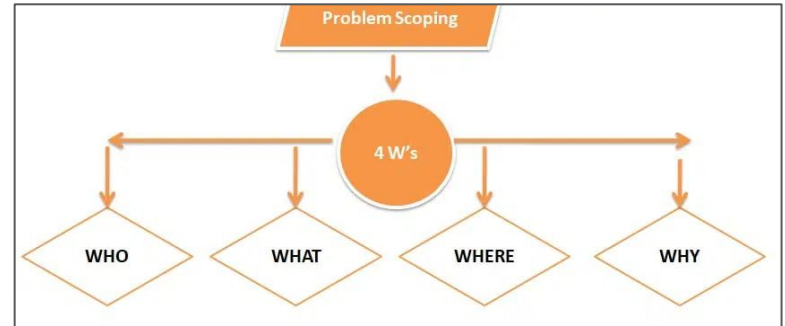


4Ws Canvas in AI Project Cycle

Stage 1: Problem Scoping

What is 4Ws Canvas?

- The 4Ws Canvas is a framework used to clearly understand a problem.
- It helps us answers four important questions:
 - WHO
 - WHAT
 - WHERE
 - WHY



Used in Problem Scoping stage.

W1 – WHO

WHO refers to:

- Who is affected by the problem?
- Who are the stakeholders?
- Who will benefit from the solution?



Examples: Students, teachers, parents, customers, drivers, etc.

W2 – WHAT

WHAT refers to:

- What is the exact problem?
- What is happening?
- What needs to be improved?

A clear and specific problem statement is required.



W3 – WHERE

WHERE refers to:

- Where does the problem occur?
- In which location or situation?



Examples:

School campus, Classroom, Road outside school, Online platform.

W4 – WHY

WHY refers to:

- Why is this problem important?
- Why should we solve it?
- What happens if we ignore it?

This explains the impact of the problem.



Example of 4Ws Canvas

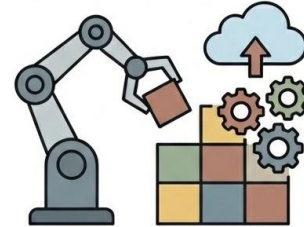
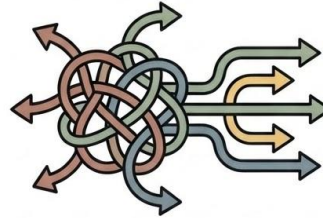
- Example Problem:
Traffic congestion near school
- **WHO** → Students, parents
WHAT → Traffic congestion
WHERE → Outside school gate
WHY → Causes delay and safety risk



Why 4Ws is Important

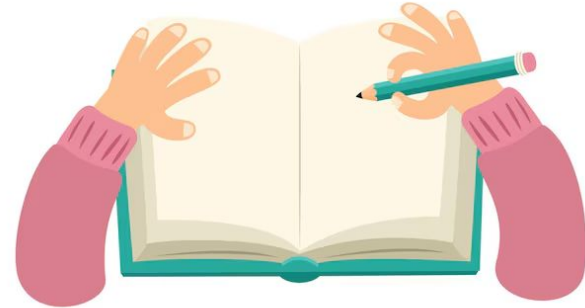
- Clarifies the problem
- Identifies stakeholders
- Prevents confusion
- Helps build better AI solutions

A well-defined problem leads to a successful AI model.



Activity – Fill the 4Ws for Pest management problem

- Fill the details for:
 - Who:
 - What:
 - Where:
 - Why:



Activity – Fill the 4Ws for Pest management problem

- Fill the details for:
 - **Who:** Farmers
 - **What:** Cotton crops got infected with pest - Pink Ballworm
 - **Where:** On the crops in the field
 - **Why:** To create an AI-enabled app that aids farmers in:
 - Determining the correct amount of pesticides
 - Knowing the right time to spray pesticides
 - Increase in the profit share of the farmers

Activity – Fill the 4Ws

Choose one problem:

- Excessive mobile phone usage
- Plastic waste in school
- Low attendance.

Students fill:

WHO, WHAT, WHERE, WHY



Activity – Fill the 4Ws

- **Answer:**
- 4Ws for Excessive mobile phone usage:
 - WHO: Students
 - WHAT: Students are using mobile phones too much, especially for games and social media.
 - WHERE: At home, in classrooms, and during study time
 - WHY: It affects students' focus, reduces study time, and can lower academic performance.



Activity – Fill the 4Ws

- **Answer:**
- 4Ws for Plastic Waste in School:
 - WHO: Students, teachers, and school staff
 - WHAT: Plastic waste such as bottles, wrappers, and bags is not disposed of properly.
 - WHERE: School campus, classrooms, playground, and cafeteria.
 - WHY: It causes pollution, makes the school environment dirty, and harms the environment.



Activity – Fill the 4Ws

- **Answer:**
- 4Ws for Problem 3: Low Attendance:
 - WHO: Students and teachers
 - WHAT: Many students are absent from school frequently.
 - WHERE: School classrooms.
 - WHY: Students miss important lessons, which affects their learning and academic performance.



Key Takeaway

- ✓ 4Ws Canvas is a tool used to clearly define a problem.
- ✓ It is used in the Problem Scoping stage of the AI Project Cycle.
- ✓ The four components are:
 - **WHO** – Identifies the stakeholders affected by the problem.
 - **WHAT** – Clearly defines the actual problem.
 - **WHERE** – Identifies the location or situation where the problem occurs.
 - **WHY** – Explains the importance and impact of the problem.

Key Takeaway

- ✓ Answering all four questions helps avoid confusion.
- ✓ A well-defined problem increases the chances of building an effective AI solution.
- ✓ 4Ws helps move from a vague idea to a clear problem statement.

A top-down view of a desk with a light blue background. In the top left, a silver laptop is partially visible. Next to it is a white mouse. Below the mouse, a pair of tortoiseshell glasses lies on the surface. In the bottom left corner, there is a dark brown leather notebook with a white pen resting on it.

THANK YOU



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